AA-F-700B
October 31, 1975
SUPERSEDING
Fed. Spec. AA-F-700A
November 20, 1968

FEDERAL SPECIFICATION

FROZEN FOOD CABINETS, MECHANICALLY REFRIGERATED, HOUSEHOLD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for self-contained, low temperature, electrically operated, mechanically refrigerated cabinets for storing frozen food.

1.2 Classification.

1.2.1 Types, sizes, grades and classes. The frozen food cabinets covered by this specification shall be of the following types, sizes, grades, and classes as specified (see 6.2).

Type I - Upright cabinet.

Size 1 - 11.0 cubic feet minimum capacity.
Size 2 - 17.0 cubic feet minimum capacity.

Type II - Chest cabinet - 18 cubic feet.

Grade A - Outer shell - Baked enamel on steel or on aluminum.

Inner liner - Aluminum, stainless steel, baked enamel on steel, or porcelain enamel on steel or plastic.

Grade B - Outer shell - Porcelain enamel on steel.

Inner liner - Stainless steel or porcelain enamel on steel or plastic.

Grade C - Outer shell - Stainless steel.

Inner liner - Stainless steel.

Grade D - Outer shell - Aluminum.

Inner liner - Aluminum.

Class 1 - Not radio interference suppressed.
Class 2 - Radio interference suppressed.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

J-C-175 - Cable Assembly Power, Electrical (3-Wire, 3 Prong, Grounding Plug Connector for 125/250 Volt Equipment).
BB-F-1421 - Fluorocarbon Refrigerants.

Federal Standards:

FED-STD-123 - Marking for Shipment (Civil Agencies).
FED-STD-151 - Metals; Test Methods.

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-R-12323 - Refrigerators and Related Equipment, Packaging and Packing of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health, Education and Welfare:

Federal Food, Drug, and Cosmetic Act as Amended Federal Regulations for Contact with Food.

(Applications for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS), Publications:

Standard For Devices to Permit the Opening of Household Refrigerator Doors from the Inside.

H 28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Director, National Bureau of Standards, Washington, DC 20234.)

American National Standards Institute (ANSI) Standards:

B38.1 - Food Storage Volume and Shelf Area of Automatic Household Refrigerators, Method of Computing.
B38.3 - Methods of Rating and Testing Home Freezers.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10016.)

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Underwriters' Laboratories, Inc. (UL), Publications:

No. 250 - Household Refrigerators and Freezers.

Electrical Appliance and Utilization Equipment List.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio St., Chicago, IL 60611; 1285 Walt Whitman Road, Melville, L.I., NY 11746; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample for first article test and approval (see 4.3 and 6.3).

3.2 Standard product. The frozen food cabinets delivered under this specification shall be the current standard product of the supplier, except for changes necessary to meet specification requirements.

3.3 Codes and standards. Frozen food cabinets shall conform to the applicable requirements of UL Standard No. 250 and the Federal Food, Drug, and Cosmetic Act, (Food additive resulting from contact with containers or equipment and food additive otherwise affecting food, Section 121.2514 - Subpart F, resinous polymeric).

3.3.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the frozen food cabinets he proposes to supply under this specification meet the applicable requirements of UL Standard No. 250 and the Federal Food, Drug, and Cosmetic Act. (Food additive resulting from contact with containers or equipment and food additive otherwise affecting food, Section 121.2514 - Subpart F, resinous polymeric).

3.3.1.1 UL. Acceptable evidence of meeting the applicable requirements of UL shall be the UL label, listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the refrigerator has been tested and conforms to the applicable requirements of UL Standard No. 250.

3.3.1.2 Federal Food, Drug, and Cosmetic Act. Acceptable evidence shall be a certification from a recognized independent testing laboratory.

3.4 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for his standard commercial frozen food cabinets provided the completed items comply with all provisions of this specification.

3.4.1 Plastics. All plastics used in the frozen food cabinet shall not impart odor or taste to foods or liquids (see 4.6.1.1). Plastics shall not craze, crack, or permanently distort from temperature or under conditions of normal use (see 4.6.1.2.1), during storage in temperatures ranging from 0 to 140°F. (see 4.6.1.2.2), or due to thermal shock (see 4.6.1.2.3). When foamed-in-place polyurethane insulation is used, all plastics in direct contact with the insulation shall be ABS (acrylonitrile, butadiene, and styrene).

3.5 Design and construction. The frozen food cabinet shall consist essentially of a cabinet and an automatic, self-contained, refrigeration system. The refrigeration system shall consist of a compressor, air cooled condenser, liquid control, evaporator, and an electric motor and motor controls, starters, protectors, and accessories. The condensing unit shall be a hermetic type, thermally or mechanically sealed. The refrigeration system shall be designed to conform to type 12 or 22 of BB-F-1421. The frozen food cabinets shall be capable of operating in ambient temperatures ranging up to 110°F. and shall be capable of safe storage and transportation without permanent impairment in air temperatures as high as 140°F. with no solar radiation and in air temperatures as low as 0°F (4.6.1.2.2 and 4.6.1.2.3).

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3.5.1 Dimensions and storage capacity. The overall dimensions and the storage capacities of each complete frozen food cabinet shall be as follows:

Type	Size	Storage Capacity (cubic feet) (minimum)	Dimensions (inches) (maximum overall) Width	Depth	Height
I	1	11.0	35-1/2	32	71-1/2
	2	17.0	37	36	73
II		18.0	57	29	39

Storage capacities shall be determined as specified in 4.6.2. If door shelf space is included in the storage capacity calculations for the freezer, the temperatures of the door shelf areas, as determined by temperature sensing devices located in the door shelf space, shall meet the requirements of this specification (see 3.5.2). If any space does not meet the performance requirements of 3.5.2, the space shall not be capable of the storage of food and shall not be included in the storage capacity calculation. The frozen food cabinet shall pass through a 30-inch wide door opening. The door and other easily removed parts may be removed to permit passage through 30-inch wide door opening.

3.5.2 Performance. When operating in a 100°F. ambient temperature, the refrigeration system shall be capable of maintaining an average food temperature of 0°F., or lower, without any food temperature exceeding 10°F. when loaded with 70 percent of the established storage volume of the cabinet and with a compressor running time not to exceed 90 percent. The refrigeration system shall also be capable of lowering the empty cabinet temperature from 110° to 0°F. in not more than 8 hours (see 4.6.5). When operating in a 90°F. ambient temperature, the refrigeration system shall be capable of maintaining a temperature of 0°F., or lower, within the cabinet with no load. The compressor running time shall not exceed 80 percent for no load and storage load operation in the 90°F. ambient temperature (see 4.6.5). Frozen food cabinets required for 50 hertz (Hz) electric current operation shall meet the above operating temperatures and performance requirements when operating on a 50 Hz electric current at rated voltage, except that an empty cabinet temperature of 100° to 0°F. in not more than 8 hours shall be required. Cabinets for 50 and 60 Hz electric current operation shall not cycle on the motor protector when tested at 90 and 110 percent of rated voltage (see 4.6.5).

3.5.3 Defrosting. Unless otherwise specified (see 6.2), defrosting shall be accomplished either automatically or manually.

3.5.3.1 Automatic defrosting. Frozen food cabinets with automatic defrosting shall be manually initiated or automatically initiated with automatic reset and shall not require the removal of food during the defrosting period. The defrosting operation shall not raise the average temperature of the frozen food more than 5°F. when the frozen food compartment is loaded with food to 70 percent of its established storage capacity. Temperature increases shall be based on the highest average temperature observed during the defrosting period and the average temperature observed just prior to initiation of defrost (see 4.6.3).

3.5.4 Cabinet. The design and fabrication of the refrigeration system and the cabinet shall prevent water vapor or water passing into the insulated space from the outside or water entering the insulated space from the inside during operation, storage, cleaning, or defrosting.

3.5.4.1 Leveling devices. Adjustable glides or leveling legs shall be provided for at least the two front corners of the frozen food cabinet.

3.5.4.2 Stability. Leveling devices and legs shall provide stability to the type I cabinet to prevent tipping when the cabinet is loaded to 70 percent of its established storage capacity, the door shelves loaded to capacity with material having a density of 35 pounds per cubic foot, and the door opened to 90° with a concentrated weight of 50 pounds suspended from the door handle (see 4.6.4.1).

3.5.4.3 Door or lid. Gaskets shall provide continuous elastic seal around the door and the method of attachment shall provide easy gasket replacement. Unless otherwise specified (see 6.2), doors on type I cabinets shall be right hand. Doors shall comply with NBS Standard for Devices to Permit the Opening of Household Refrigerator Doors from Inside.

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3.5.4.3.1 Door or lid inner liner. Door inner liner shall be fabricated of the material specified for the cabinet inner liner (see 3.5.4.4), or shall be styrene alloy (high impact polystyrene), ABS plastic (acrylonitrile, butadiene, and styrene), or copolymer plastic.

3.5.4.3.2 Antisweat device. Antisweat devices will be allowed around the periphery of the door opening as required.

3.5.4.3.3 Hardware.

3.5.4.3.3.1 Door or lid latch and strike assembly. The door latch and strike assembly, where used, shall be fabricated of steel or malleable iron, copper-nickel-chromium plated; or cast or forged brass, nickel-chromium-plated; or of high pressure die cast zinc, copper-nickel-chromium plated. Nylon contacts may be used in the latch and strike assembly. The latch and strike assembly with handle shall be adjustable to compensate for a 10 percent minimum reduction in original gasket thickness or shall be designed to automatically provide sufficient closing force to effect a gasket seal required over a range of gasket thickness reduction ranging from 0 to 10 percent. Original gasket thickness shall be measured at the latch or handle side of the door or lid with the door or lid closure as adjusted to effect the gasket seal required.

3.5.4.3.3.2 Hinges. Exposed hinges shall be steel or brass, copper-nickel-chromium-plated, or stainless steel. Concealed or semiconcealed hinges may be steel finished to match the exterior of the frozen food cabinet. Hinges shall be adjustable to compensate for at least a 10 percent reduction of compressible height of original gasket thickness on the hinge side, except that where magnetic gaskets are used, adjustment is not required. Hinges on type II cabinets shall be provided with manufacturer's standard counter-balance device which shall hold the lid in any open position between 5° and 70° and open to 90°. The device shall counterbalance the lid so it may be raised with maximum effort of 5 pounds (see 4.6.4.2).

3.5.4.3.3.3 Door or lid closure. The latch and strike assembly, or other means used for holding the door or lid in a normally closed position and the hinges shall hold the door or lid closed with sufficient force to effect a gasket seal. Gaskets shall fit the cabinet evenly and shall provide an air tight, leakproof joint when the door or lid is normally closed (see 4.6.4.2).

3.5.4.4 Cabinet inner liner. The inner liner of the frozen food cabinet shall be fabricated of material according to the grade of frozen food cabinet as indicated in 1.2. Baked enamel shall be in accordance with the manufacturer's standard practice and, in addition, shall comply with finish requirements of Federal Food, Drug, and Cosmestic Act (see 3.3.1.2). The interior shall be free from voids, cracks, or imperfections which hinder cleaning. Unless otherwise specified (see 6.2), the color of baked enamel and porcelain enamel finishes shall be white. Inner liners may be sectional but all joints shall be sealed tight and fastened to assure structural rigidity. If seams are welded, the exposed surfaces shall have all discoloration removed. The insulation side of sheet steel inner liners, reinforcement, and stiffeners shall be cleaned, phosphatized, and given a prime coat of paint. All corners shall be designed for ease in cleaning. Sheet steel for baked enamel or porcelain finish shall have a thickness of 0.0210 inch or greater; stainless steel shall be polished and shall have a thickness of 0.022 inch or greater; and aluminum alloy shall have a thickness of 0.0295 inch or greater. Aluminum alloy sheet with a thickness of 0.0225 inch may be used when embossed liners are furnished.

3.5.4.4.1 Breaker strips. Breaker strips of thermoplastic material, thermosetting plastic, natural rubber, or synthetic rubber shall be installed to joint the outer shell and inner liner around exterior door or lid openings on type I cabinets. Joints between the cabinet inner liner and the bottom breaker strip and the lower 4 inches of the side breaker strips shall be sealed to prevent liquid spillage or defrost water from entering the insulation space. Breaker strips shall not be damaged or displaced by thermal shock (see 4.6.1.2.3). Joints in breaker strips shall be smooth and cleanable; edges of strips shall be eased or rounded; and the entire installation shall provide a smoothly finished appearance. The bottom breaker strip shall not show any cracks or crazing when given the 2-pound, 1/2-inch spherical radius dart or tup impact test as specified in 4.6.1.2.4.

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3.5.4.5 Interior cabinet accessories.

3.5.4.5.1 Type I cabinet. The frozen food cabinet shall be equipped with not less than three full-width shelves. When the bottom of the cabinet liner is not a refrigerated surface or when a swing out or slide out basket is not furnished for the bottom of the cabinet, a floor rack to support the frozen food shall be provided. The floor rack shall cover the entire floor of the cabinet, including horizontal and slanted sections, and shall provide 3/4-inch minimum clean airspace below the bottom of the rack. Permanently mounted, full width, refrigerated plates, the bottom of suspended containers, or the floor of any fixed full-width compartment shall be considered a shelf. Any shelf within 3 inches of the floor or top of the cabinet inner liner shall not be considered a shelf. Shelves other than refrigerated plate shelves shall be removable and shall be designed to prevent loosening or tilting when in use. Shelves, as installed, shall not deflect more than three-eights of an inch upon application of an 80-pound weight (see 4.6.4.3). Juice container racks, door shelves, and similar containers shall be readily cleanable or removable for cleaning without the use of tools, and without removing the door liner.

3.5.4.5.2 Type II cabinets shall be provided with the manufacturer's standard basket constructed of materials specified herein.

3.5.4.5.3 Accessories and finish. Metal shelves, baskets, and floor racks shall be constructed of passivated stainless steel; steel tube, steel wire, or steel tube on sheet, either copper-nickel-chrome plated, or electroplated zinc plus protective coating; anodized aluminum; or baked enamel on phosphatized steel. In addition to the materials stated above, juice container racks, door shelves, and similar special containers may be constructed of styrene plastic or equivalent as shown by certified comparative test data. When exposed to a salt spray test for 96 hours, metal shelving shall show not more than 3 visible spots of attack in 12 square inches (see 4.6.4.4). When baked enamel on steel is used, the steel shall be phosphatized and the baked enamel shall be in accordance with the manufacturer's standard practice. The baked enamel finish shall be free of runs, sags, streaks, seeds, orange peel, dusting, mottling or peeling.

3.5.4.6 Outer shell. The outer shell shall be fabricated of aluminum alloy having a minimum thickness of 0.040 inch with a standard mill finish without stains or discoloration when the aluminum is not coated with an organic finish; steel finished in porcelain (vitreous) enamel, baked acrylic enamel, or baked alkyd-amine enamel; or stainless steel, according to the grade of frozen food cabinet as indicated in 1.2. Aluminum alloy finished in baked alkyd-amine enamel may be used for the outer shell of the cabinet when a baked alkyd-amine enamel exterior finish on steel is specified, provided that these materials are normally used by the supplier in his standard product. The porcelain enamel shall not be discolored, chipped, blistered, lumpy, fish scaled, or have burn-off crazing, beads of enamel or glaze, lumps or scumming. The baked enamel finish shall be free of runs, sags, seeds, streaks, orange peel, dusting, mottling or peeling. When exposed to a salt spray for 96 hours (see 4.6.4.4), the outer shell materials shall show no corrosion. The insulation side of steel outer shells, reinforcement, and stiffeners shall be cleaned, phosphatized, and given a prime coat of paint. Unless otherwise specified (see 6.2), outer shells finished in porcelain or baked enamel shall be white. Joints that are not welded airtight shall be sealed to prevent moisture infiltrating the insulation. The sealing compound shall make a permanent watertight seal. Openings in the shell for entrance of tubing shall be mechanically sealed to prevent moisture infiltration.

3.5.4.7 Insulation. The cabinet and door or lid shall be insulated with fiberglas, expanded polystyrene, slab polyurethane, or foamed-in-place polyurethane insulation (see 3.4.1). The insulation thickness shall be sufficient to prevent all the exterior surfaces from being more than 10°F. below the room ambient temperature when the frozen cabinet is operating (see 4.6.4.5).

3.5.5 Refrigeration system.

3.5.5.1 Evaporator. The evaporator shall be forced air or gravity type and shall be tube and fin; tube and wire; tubing bonded, welded, or crimped to the underside of the shelves; embossed plates; tubing bonded to the insulation side of the inner liner; or shall be a combination of these designs provided that the evaporator materials and design have been in commercial use for at least one year other than experimental service. Plates may be single or double embossed, and may be formed to refrigerate more than one shelf. Bonding of tubing to the inner liner shall provide maximum possible thermal contact.

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3.5.5.1.1 Defrost water. Defrost water on automatic defrost systems shall be carried out of the cabinet to evaporating apparatus. Automatic defrost systems shall have provisions to prevent defrost water from dropping on the food. Tubes, pipes, or baffles used on automatic or manual defrost systems, to collect or carry defrost water, shall have corrosion-resistant surfaces and shall be readily cleanable or shall be self-cleaning.

3.5.5.2 Condenser. The condenser shall be forced air or gravity type and shall be finned tubing, tube on plate, embossed plates, or wire and tube, provided that the materials and design have been in commercial use for at least one year of service other than experimental service. When spacing of the cabinet from the wall is required to assure proper ventilation, spacer bolts, a duct, or other means shall be provided to assure proper spacing. Portions of condensers most likely to become blocked with dust, such as finned tubes in the air stream of a fan, shall be readily accessible for cleaning.

3.5.5.3 Charging of refrigeration system. The refrigeration system shall be charged with refrigerant (see 3.5), and oil recommended by the manufacturer. After charging, the refrigeration system shall not freeze up due to moisture in the refrigerant when the frozen food cabinet is operated (see 4.6.6.1). The moisture content shall not exceed 10 parts per million (ppm) for R-12 and 25 ppm for R-22 (see 4.6.6.2).

3.5.6 Electrical requirements. Unless otherwise specified (see 6.2), frozen food cabinets shall be designed for operation on a nominal 120-volt, 60 Hz single-phase, electrical system, and shall be grounded by a cord conforming to type I of J-C-175, except for length. The cord shall be not less than 5 feet long from the point of attachment at the back of the cabinet. Any special transformers required to meet specified voltage requirements shall be furnished with each frozen food cabinet.

3.5.7 Protection against electrolytic or galvanic corrosion. Frozen food cabinets shall be constructed to provide protection against electrolytic or galvanic corrosion. Joints between dissimilar metals, which may become wetted in service, shall be protected against electrolytic or galvanic corrosion by proper selection of materials, plating, isolation, insulation, paint, area relationship, gaskets, or other means.

3.5.8 Operation. The frozen food cabinet shall cycle properly; operate according to control settings; mechanical parts shall be in working order; and there shall be no refrigerant leaks (see 4.6.1.2.3 and 4.6.7).

3.6 Marking for identification. The manufacturer's name that normally appears on the outside of his product and any other standard name plate data shall be placed on all frozen food cabinets according to the manufacturer's standard practice.

3.7 Wiring diagram. A wiring diagram shall be glued or otherwise affixed to the back of the cabinet, the machine compartment cover, or the condenser duct. The diagram shall show each electrical component by sketch, outline, or symbol, and shall include the name of each component. The wiring diagram shall show the wiring to all components and any coding used for identification of wires.

3.8 Electromagnetic compatibility. Class 2 frozen food cabinets shall be designed and equipped for electromagnetic compatibility in accordance with the class IIB requirements of MIL-STD-461 (see 4.6.8).

3.9 Workmanship. Welding and brazing shall be complete, uniform, and properly fused having no burn holes, slag, inclusions, scale, or flux deposits, and shall not be cracked, fracture, or undercut. Soldering shall be complete clean, and adherent, and shall not have any pinholes. Bolts, nuts, screws, studs, and other types of threaded fasteners when used, shall not be broken, stripped, fractured, or loose, and shall have lock-washers or shall be a self-locking type.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article. When a first article is required, the frozen food cabinets shall be examined for defects listed in table I, for dimensions and coating thickness specified herein, and tested as specified in 4.6.1.2 through 4.6.4.3, 4.6.4.5, 4.6.5, 4.6.6.2, 4.6.7 and when required 4.6.8.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Plastics. The plastic test shall be performed as specified in 4.6.1.1. The inspection lot shall be the amount of each type of plastic offered for inspection at one time. Any nonconformance shall be cause for rejection of the lot.

4.4.1.2 Material for outer shell. Materials for the outer shell shall be tested as specified in 4.6.4.4. Three specimens, 3 by 6 inches and representative of the end item outer shell, shall be used for the test. Any nonconformance shall be cause for rejection.

4.4.1.3 Shelving. Shelving shall be tested as specified in 4.6.4.4. The inspection lot shall be the number of shelves offered for inspection at one time. Any nonconformance shall be cause for rejection of the lot.

4.4.2 In-process inspection.

4.4.2.1 Process examination. Examination shall be made of the following processes to establish conformance to specified requirements. Whenever non-conformance is noted, correction shall be made to the affected items and process.

(a) Sealing of openings in cabinet shell against moisture infiltration (see 3.5.4 and 3.5.4.6).
(b) Inner liner joints sealed (see 3.5.4.4).
(c) Corrosion-resisting treatment of sheets, reinforcement, and stiffeners for inner liner and outer shell (see 3.5.4.4 and 3.5.4.6).
(d) Installation of proper insulation (see 3.5.4.7).
(e) Bonding of tubing to the inner liner, when applicable (see 3.5.5.1).
(f) Protection of joined dissimilar metals against corrosion (see 3.5.7).

4.4.3 End item inspection. The inspection lot shall consist of all the frozen food cabinets of one type, size, grade, and class offered for inspection at one time. The sample unit shall be one frozen food cabinet.

4.4.3.1 Visual examination. Examination shall be made of the frozen food cabinet for the defects listed in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 4.0 for major and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished, plated, or coated (where required).	X	
	Type or color of finish not as specified.		X
	Not ABS plastics in contact with foamed insulation.	X	

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TABLE I. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Design, construction and workmanship general (applicable to all components and assemblies)	Fractured, split, punctured, torn, dented, creased, bowed, sprung, or malformed.	X	
	Part missing, misplaced, loose, or not in proper alignment, or not type specified.	X	
	Functioning component that is inoperative or will not operate as intended.	X	
	Adjustable assembly that cannot be adjusted, or is not properly adjusted to perform the function intended (condition cannot be readily rectified).	X	
	Not properly adjusted but can be readjusted.		X
	Component not accessible for servicing where required.		X
	Welding and brazing missing, having burn holes, cracked, fractured, or not fused.	X	
	Welding and brazing having slag, inclusions, is undercut, not uniform, with scale or flux deposits not removed.		X
	Soldering missing, not adherent.	X	
	Soldering not clean (flux or flux residue not removed); pinholes in solder or incomplete.	X	
	Bolts, nuts, screws, studs, and other types of threaded fasteners missing, broken, stripped, fractured, or loose.		X
	Bolts, nuts screws, studs, and other types of threaded fasteners having lockwashers missing where required.		X
Wiring diagram	Missing, incomplete, not legible, not affixed in the prescribed manner.		X
Rivets (when applicable)	Missing, loose, broken, not peened, or insufficiently peened.		X
Marking for identification	Missing, incomplete, not legible.		X

4.4.3.2 Dimensional examination. Examination shall be made on the frozen food cabinets for defects in dimensions. Any dimensions not within the requirement specified in 3.5.1 shall constitute a defect. The inspection level shall be S-2 with an AQL of 6.5, expressed in terms of defects per hundred units.

4.4.3.3 End item testing. Each end item shall be tested as specified in 4.6.6.2, when applicable, and 4.6.7. Any nonconformance shall be cause for rejection of the lot. One representative production unit (see 6.6), shall be tested as specified in 4.6.1.2 through 4.6.4.3, 4.6.4.5, 4.6.5 and when required 4.6.8. Any nonconformance shall be cause for rejection. When a combination liquid and moisture indicator is furnished with each unit, one production unit from the initial lot shall be tested as specified in 4.6.6.1. Any nonconformance shall be cause for rejection of the lot. When a combination liquid and moisture indicator is not furnished with each unit, the test specified in 4.6.6.1 shall be performed using an inspection level of S-1 and an AQL of 6.5, expressed in terms of defects per hundred units.

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4.4.4 Inspection of preparation for delivery. An examination shall be made to determine that preparation for delivery as specified in section 5 is in conformance to MIL-R-12323.

4.5 Certification examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.6 Tests.

4.6.1 Plastic and cabinet tests.

4.6.1.1 Odor and taste test. Plastic shall be given the following odor and taste test to determine compliance with the odor and taste requirements of 3.4.1. Place the plastic to be tested with approximately 5 cc of distilled water in a 16-ounce, wide-mouth, glass jar. Place a beaker containing 100 to 150 cc of distilled water in the jar so that the water does not touch the plastic being tested. The jar shall be closed tightly. Place 100 to 150 cc of distilled water in a second closed glass jar. The two jars shall be kept at 45° to 50°F. for 24 hours. At the end of this period an examination shall be made for odor or taste difference between the two waters. Containers of other shapes, capacities, and types may be substituted for the containers described, if odd shaped pieces of plastic are tested, provided the intent of the odor and taste test is met.

4.6.1.2 Normal use and storage tests.

4.6.1.2.1 Normal use test. The assembled frozen food cabinet shall be permitted to come to steady state with the interior cabinet temperature at the lowest value obtainable with continuous operation of the refrigerating unit in an ambient temperature of 100°F. to determine compliance with the plastic normal use requirement of 3.4.1.

4.6.1.2.2 Storage test. The inoperative frozen food cabinet without storage load and door open shall be permitted to come to thermal equilibrium in ambient temperatures of 0° and 140°F. to determine compliance with the storage requirements of 3.4.1 and 3.5.

4.6.1.2.3 Thermal shock transportation tests. The cabinet shall be placed in an ambient temperature of 140°F. for a period of 12 hours. During the last hour of the 12 hour period, the frozen food cabinet shall be subject to the following two tests:

(a) One side of the frozen food cabinet shall be raised 3 inches off a level concrete floor by a single support which can be withdrawn to allow a free drop. This procedure shall be repeated allowing each of the four sides to drop.

(b) The cabinet door shall be opened 90° and slammed closed 10 times. At the end of the 12 hour period, the frozen food cabinet shall be transferred, as quickly as practical, to room maintained at 0°F. The freezer shall remain in this room for 12 hours. During the last hour of this 12-hour period, the cabinet shall be resubjected to the two tests (drop and door slamming) described above. During and after these tests, the frozen food cabinet shall be examined for compliance with the thermal shock requirements of 3.4.1, 3.5, and 3.5.4.4.1. After the frozen food cabinet has returned to room temperature, it shall be operated for 2 hours and examined for compliance with the normal use and operating requirements of 3.4.1 and 3.5.8. The test of 4.6.1.2.2 may be run concurrently with the test in this paragraph.

4.6.1.2.4 Bottom beaker strip impact test type I. The storage compartment temperature shall be in accordance with the no-load test conditions of ANSI B38.3. A 2-lb, 1/2-inch spherical radius, hardened steel dart or tup shall be dropped from a minimum height of 6 inches onto the bottom breaker strip of the frozen food stroage compartment as assembled in the cabinet. The dart or tup shall impact the breaker strip once after each drop. Three noncoincident points at random shall be tested and the test shall be carried out within 30 seconds after opening the door. After completion of the tests, the beaker strip shall be examined for cracks or crazing.

4.6.2 Storage capacity test. To determine compliance with the storage capacity requirements of 3.5.1, the following determination shall be made:

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4.6.2.1 Type I.

A. The total frozen food stroage volume shall be the sum of the main compartment storage volume and the door storage volume.

1. Measurement of main compartment storage volume.

(a) Each shelf shall be measured separately.
(b) Using the front edge of the shelf as a reference, measure the depth of the compartment to the inside of the back well.
(c) Measure the width of the compartment from wall to wall.
(d) Measure the height of the space from the upper surface of one shelf to the under side of the refrigerated surface or the under side of the shelf above.
(e) Multiply the width, depth, and height obtained in accordance with items 1 (b), (c), and (d), with applicable adjustments, to obtain the frozen food storage volume of each shelf.
(f) If any shelf contains a basket, the inside volume of the basket shall be the frozen food storage volume of that shelf.
(g) Any compartment or space having a vertical height of 2 inches or less shall not be included in the calculations.

2. Measurement of door storage volume.

(a) Each shelf shall be measured separately.
(b) For shelves of parallel piped shaped, measure the three major dimensions of each shelf. Disregard fillets in corners unless the radius of the fillet is greater than 1-1/4-inch. Disregard taper of the space unless it is greater than 1/2 inch on a side. If the above limits of fillet radius and taper are exceeded, use mean dimensions in each direction. In measuring the depth of a door shelf, no space closer than 3/4 inch to the front edge of a shelf in the main storage compartment, as determined for a closed door position, shall be included as door storage volume. Multiply height, depth, and width measurements determined in accordance with this paragraph to determine the frozen food storage volume of each shelf in the door.
(c) The volume of shelves designed especially to fit cylindrical cans of frozen food shall be based on the actual volume of a full load of the largest size can that the shelf can accommodate.

B. In lieu of making measurements as specified above for capacity determination, the minimum storage capacity may be established by packing the freezer shelves, basket, and door shelves with standard size frozen food packages and cans. Packages on cabinet shelves shall not extend over the edge of such shelves, and packages shall not be placed under or around the exterior of baskets nor in cabinet spaces on shelves with less than 2 inch vertical clearance above the shelves. All packages shall be filled solid to prevent compressing or forcing out of shape so as to result in volume change. Wood or other solid materials, cut and shaped to represent standard frozen food packages, may be used at the supplier's option. It shall be observed that the size 1 freezer holds a minimum of 11 cubic feet of standard size packages, and the size 2 freezer holds a minimum of 17 cubic feet of standard size packages.

NOTE: For cabinets where door shelves are not provided, and for cabinets where door shelves are provided but are not included in the capacity calculation because of high temperature in the door shelf area, the "depth" measurement shall be made from the face of the liner at the back of the cabinet to the front edge of the cabinet shelves.

4.6.2.2 Type II.

A. Storage capacities shall be determined as outlined in ANSI B38.1, except that the volumes occupied by inclosed unusable spaces within the freezer main compartment or unusable space between the basket and freezer walls shall be deducted from the refrigerated space in determining the storage capacities.

B. In lieu of making measurements as specified above for capacity determination, the minimum storage capacity may be determined by packing the freezer compartment (baskets if applicable) with standard size food packages and cans. All packages shall be filled solid to prevent compressing or forcing the packages out of shape. Wood or other solid materials, cut and shaped to represent standard frozen food packages, may be used at the supplier's option. It shall be observed that the type II freezer holds a minimum of 18 cubic feet.

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4.6.3 Defrost tests. The frozen food cabinet with automatic defrosting shall be tested as follows to determine compliance with the defrosting requirements of 3.5.3.1.

4.6.3.1 Defrost test conditions. The cabinet shall be loaded with sealed test packages, as described in ANSI B38.3, to 70 percent of its established storage capacity with material having a density of 35 pounds per cubic foot. Commercially packaged chopped spinach may be used in lieu of the material described in ANSI B38.3 provided the spinach has a density of 35 pounds per cubic foot. Ambient conditions for the test shall be 90°F. and a relative humidity between 80 and 85 percent. The temperature control unit shall be adjusted to obtain an average cabinet air temperature of 0° ± 2°F. Electrical characteristics for operation of this test shall be the same as equipment rating.

4.6.3.2 Procedure. During the first hour of the initial 8 hours of the test, the door or lid shall be opened wide seven times per hour for 10 second periods at 8 minute intervals, and the procedure shall be repeated during the fourth and seventh hours. Following the door or lid opening schedule, the frozen food cabinet shall stabilize for the next 16 hours without doors being opened. The defrosting shall begin immediately following the stabilization period. Automatically initiated defrost systems may be prepared prior to the test to prevent defrosting during the door or lid opening and stabilization portion of the test. The defrost period shall be terminated when the temperature of the cabinet has peaked and the frozen food cabinet shall then continue to operate over a 4 hour recovery period. The temperature of the frozen food cabinet and the test packages shall be taken continuously, or at intervals of not greater than 15 minutes, starting prior to initiation of the defrost cycles and continuing throughout the recovery period.

4.6.4 Cabinet tests.

4.6.4.1 Stability test. The frozen food cabinet shall be loaded to 70 percent of its established storage capacity and the door shelves, when applicable, shall be fully loaded with material having a density of 35 pounds per cubic foot, as described in ANSI B38.3, and a concentrated weight of 50 pounds shall be suspended from the door latch and the door shall be opened to 90° F. to determine compliance with 3.5.4.2.

4.6.4.2 Door or lid seal test. The door or lid of the frozen food cabinet shall be closed on the strip of paper of the width and thickness of new United States currency and the paper withdrawn to determine compliance with 3.5.4.3.3.3. The test shall be performed at least once at each side and at the top and bottom of the door. If the configuration of the door or trim does not permit this test, a minimum 100-watt lamp shall be placed inside the frozen food cabinet and an observation for light showing between the gasket and seating face made in a darkened room. The lid on type II cabinet shall be tested for conformance to 3.5.4.3.3.2.

4.6.4.3 Shelf test type I cabinet. An 80 pound load shall be placed on a 12 by 12 inch square piece of plywood, centered on the shelf to determine compliance with the shelf requirements of 3.5.4.5.1. The plywood shall not rest on any rim of the shelf. When clear space above the shelf does not allow room for locating a compact 80 pound load, the shelf shall be mounted in a mock-up or jig to approximate its position or mounting in the cabinet.

4.6.4.4 Salt-spray test. Outer shell material (see 4.4.1.2) and metal shelving (see 4.4.1.3) shall be subjected to a salt-spray test in accordance with Method 811 of FED-STD-151 for 96 hours to determine compliance with 3.5.4.5 and 3.5.4.6, respectively.

4.6.4.5 Exterior surface temperature test. The exterior surface temperatures of the frozen food cabinet shall be tested to determine compliance with the insulation requirements of 3.5.4.7. During the performance of the test specified in 4.6.5, when the frozen food cabinet is operating and the inside temperature is at equilibrium, the entire exterior of the cabinet, including doors or lids, shall be inspected to determine that all surfaces are not more than 10°F. below the test room ambient temperature.

4.6.5 Performance tests. The frozen food cabinet shall be subjected to an average food temperature test, a storage load test, a no load test, and a no load pulldown test to determine compliance with temperature and compressor running time requirements of 3.5.2. Automatic defrost systems may be shunted out for these tests. The tests shall follow the procedures in ANSI B38.3 with modified conditions as required by this specification and with the requirement that the average food temperature test shall be made in accordance with the same procedure outlined for the modified storage load test. Refrigeration units procured for operation on a frequency of 50 Hz shall be tested by operating for 4 hours at the specified frequency and at 90 percent of the specified system voltage followed by an operating test of 4 hours at specified frequency and at 110 percent of specified system voltage to determine compliance with the motor cycling requirements of 3.5.2. The loads for the average food

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temperature test and the storage load test shall be 70 percent of the storage capacity as determined by 4.6.2. The test for no load pulldown shall be made in accordance with ANSI B38.3 with the thermal equilibrium temperature at completion of test not to exceed 2°F. Average food temperature tests shall be conducted in an ambient temperature of 100°F. The no load and the storage load tests shall be made in an ambient temperature of 90°F.

4.6.5.1 Type I. The temperature sensing devices in the interior of the frozen food cabinet shall be located in each compartment or section to provide data on average temperature and high and low temperature of each compartment or section, subject to location requirements specified hereinafter. A compartment or section is defined as a space designed for storage of food, bounded by the inner liner, the top of the immediate lower shelf, the bottom of the immediate upper shelf, and a plane passing vertically through the front edge of the shelf.

The temperature sensing devices in the storage shelves, spaces, and racks of the door shall be placed such that separate temperature data are provided for each separate shelf section, space, or rack designed for storage of food.

Temperature sensing devices are not required in any spaces within the body or door of the frozen food cabinet that are not designed for the storage of food, such as the space below a basket where a basket is provided, and the space above a top shelf plate where such plate is located to provide less than 2 inches clear vertical space between the top of the plate and the inner liner. Spaces where temperature is not recorded shall not be included in the computed storage capacity of the cabinet.

Each compartment or section in the cabinet shall have a minimum of three temperature sensing devices located on the front to back centerline as follows:

(a) One sensing device 1 inch beneath the underside of the top boundary and 1 inch from the face of the inner liner back.
(b) One sensing device at the geometric center.
(c) One sensing device 1 inch above the top of the bottom boundary and 1 inch in back of the front edge of the shelf.

In the top compartment, one additional sensing device shall be placed 2 inches beneath the underside of the top boundary and 1 inch in back of a plane projected through the front edge of the cabinet shelf.

Where a basket is provided in any compartment, three sensing devices shall be installed in the basket space as specified for a compartment, and two additional temperature sensing devices shall be placed on the bottom of the basket, at the 1/3 points of the cabinet front to back dimension.

When no basket is provided for the bottom storage space, the temperature sensing devices shall be installed as specified for compartments, and two additional temperature sensing devices shall be placed on the top of the floor rack; one will be placed at the center point and one at the rear of the floor rack on the front to back centerline of the cabinet.

Where door storage space is provided, at least one or more temperature sensing devices shall be located in each storage shelf, shelf section, space, or rack designed for frozen food storage.

When door shelves are not provided, temperature sensing devices shall be installed on the vertical centerline of the cabinet in a plane 1 inch from the face of the door liner, with one located 1 inch below the top of the cabinet liner, one at the centerpoint, and one at the bottom 1 inch above the bottom inner liner.

4.6.5.2 Type II cabinet. The temperature sensing devices in the interior of the frozen food cabinet shall be located to provide data on average temperature and high and low temperature subject to location requirements specified hereinafter.

Temperature sensing devices are not required in any spaces within the body of the frozen food cabinet that are not designed for the storage of food such as the space below the basket where a basket is provided or space containing refrigeration unit.

Temperature sensing devices shall be located inside the freezer compartment in accordance with ANSI B38.3.

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4.6.6 Refrigerant system moisture test.

4.6.6.1 Freeze-up moisture test. The frozen food cabinet shall be tested as follows to determine compliance with the moisture requirements of 3.5.5.3. The frozen food cabinet shall be connected to a source of power and operated continuously for 8 hours in a temperature of 110°F. ambient, the power shall be shut off and the unit allowed to stabilize for 8 hours in an ambient temperature 70°F. Immediately after this 8 hour period, the frozen food cabinet shall be operated continuously as above for 8 hours in an ambient temperature at 70°F. During this period, examination shall be made for evidence of refrigerant system freeze-up. When the combination liquid and moisture indicator is used, it shall indicate "dry" within the limits specified in 3.5.5.3.

4.6.6.2 Indicator moisture test. When a moisture indicator or a combination liquid and moisture indicator is furnished with each unit, the indicator shall be observed after the operation test (4.6.7). A liquid line temperature of not more than 100°F. shall be maintained when observing the indicators. The indicators shall indicate "dry" within the limits specified in 3.5.5.3.

4.6.7 Operation test. The frozen food cabinet shall be operated 30 minutes at a normal control setting to determine compliance with 3.5.8. After completion of this portion of the test, and with the cabinet temperature at 70° + 3°F., all accessible refrigerant joints shall be tested for leaks to determine compliance with 3.5.8. All refrigerant joints not accessible after assembly of the frozen food cabinet shall be tested by a means and method equivalent to the above test.

4.6.8 Electromagnetic compatibility tests. When electromagnetic compatibility is required the first article or initial production unit, as applicable, shall be tested by the supplier in accordance with test method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-461.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Each frozen food cabinet shall be preserved and packaged in accordance with the applicable requirements of MIL-R-12323.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. Each frozen food cabinet, preserved and packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of MIL-R-12323.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123. Precautionary and handling markings shall apply.

5.3.2 Military requirements. In addition to any special marking required by contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129. Precautionary and handling markings shall apply.

6. NOTES

6.1 Intended use. The frozen food cabinets covered by this specification are intended for household use to store frozen food at 0°F. in ambient temperatures up to 100°F.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, size, grade, and class required (see 1.2.1). Class 1 will be furnished if class is not specified in procurement documents.
(c) First article. Whether requirement for preproduction sample is waived (see 3.1).

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(d) Whether a specific method of defrosting is required (see 3.5.3).
(e) Whether a left hand door is required (see 3.5.4.3).
(f) Color of finish of inner liner if other than white (see 3.5.4.4).
(g) Color of porcelain or baked enamel for outer shell if other than white (see 3.5.4.6).
(h) Electrical requirements, if other than specified (see 3.5.6).
(i) Selection of applicable levles of preservation, packaging, and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Definitions. The following definitions are applicable to this specification:

Readily cleanable: Readily or easily cleanable shall mean readily accessible and of such material, finish, and fabrication that soil may be effectively removed by normal cleaning methods.

Smooth: The word smooth is used to define a surface free of pits and inclusions and having a cleanability equal to commercial grade hot roll steel free of visible scales.

6.5 The frozen food cabinets procured under this specification, which are centrally inventory managed, shall be limited to the varieties set forth in FED-STD-247. The specification may be used for the procurement of additional varieties which are not centrally procured or are approved in accordance with existing regulations.

6.6 End item tests. Performance of tests as specified in 4.6.1 through 4.6.4.3, 4.6.4.5, 4.6.5, and when required 4.6.3 should be limited to one successful demonstration of compliance to the requirements of this specification for each contract.

CUSTODIANS:

Army - GL
Air Force - 82

Review activity:

Army - MD

Preparing activity:

Army - GL

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Project No. 4110-0231

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